UNIDAD ACADÉMICA MULTIDISCIPLINARIA MANTE

DISEÑO ELECTRÓNICO BASADO EN SISTEMAS EMBEBIDOS

Actividad:

Configurar PlatformIO en Visual Studio Code

Alumnos(as):

- Castro Márquez Cristian
- García Segura América Yanely
- López Camacho Jonathan Valentín
- Navarro García Maribel
- Ramírez Bustos Joshua Romany

Maestro(a):

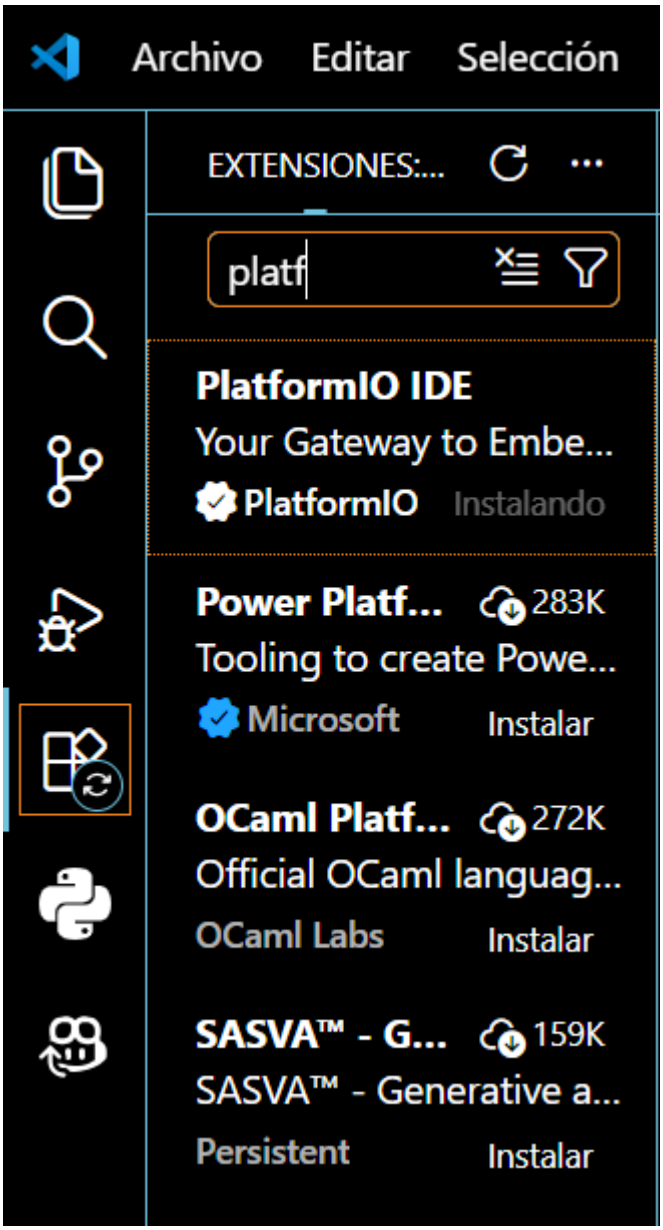
López Piña Daniel

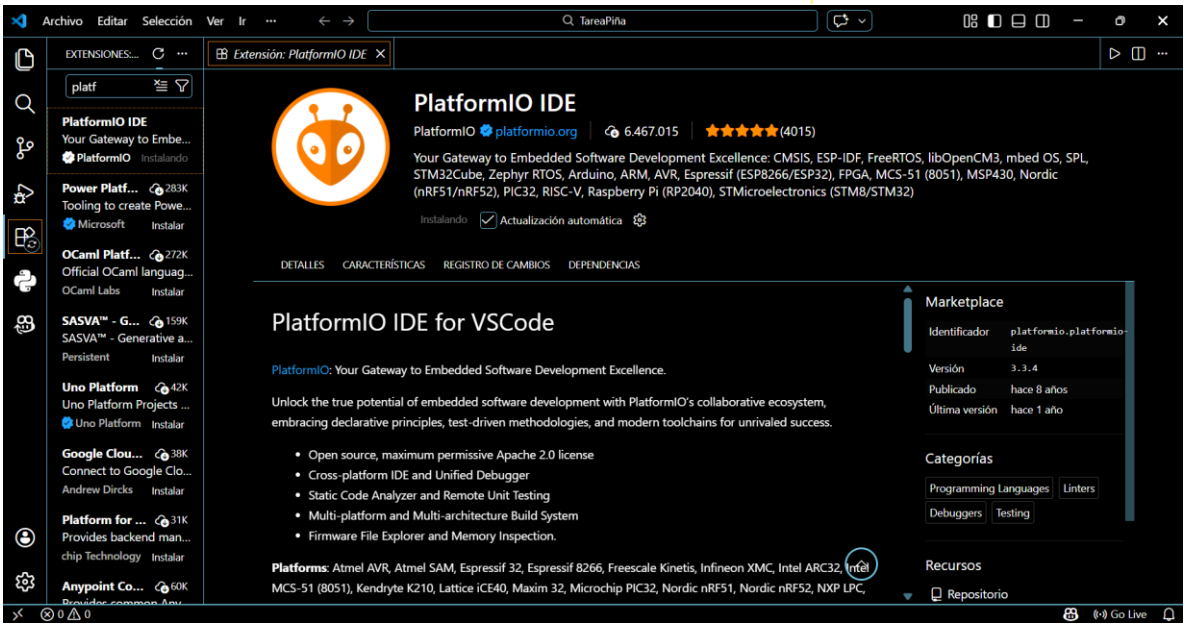
Ciudad Mante, Tamaulipas Marzo, 2026

Contenido

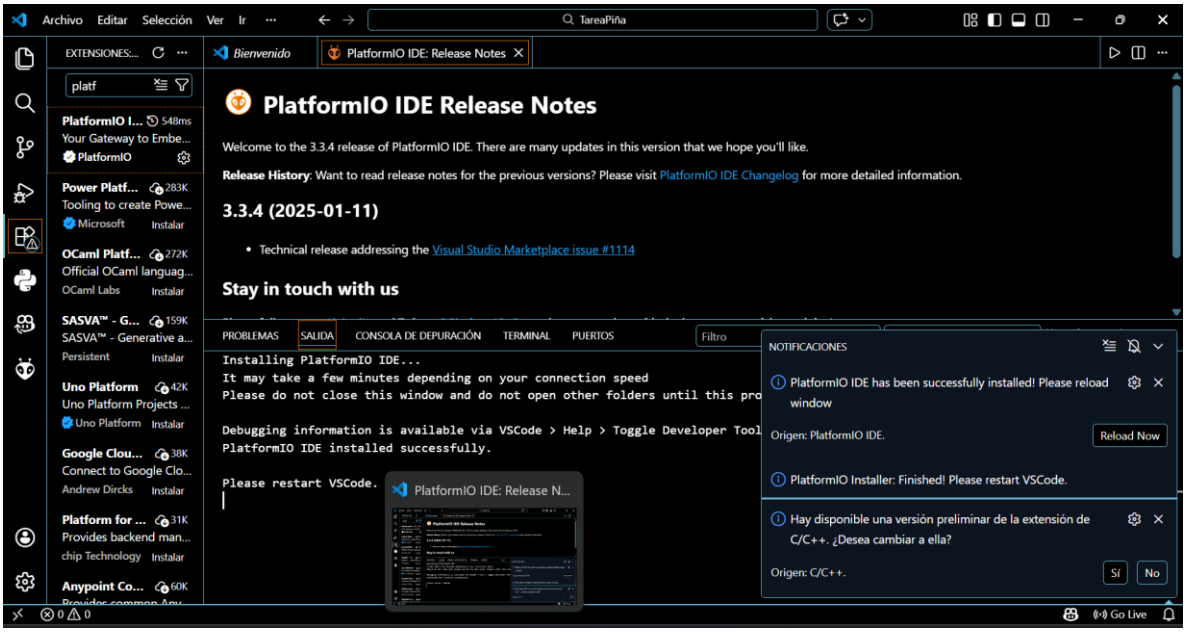
Instalación de extensión PlatformIO 3
Descarga de su núcleo (core) 4
Crear el primer proyecto 5
Anatomía de un proyecto profesional 7
Compilar y subir el código a hardware 8
Código de prueba 8
Librerías 10
Código de ejemplo en una ESP32 13

Instalación de extensión PlatformIO

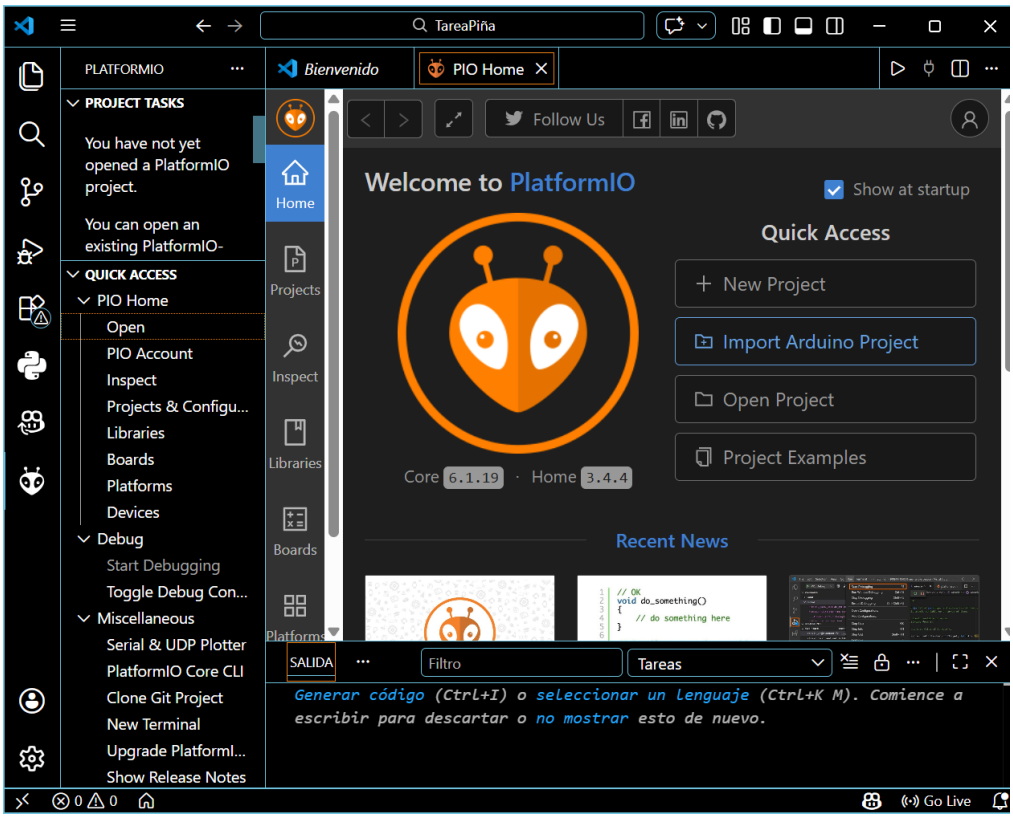


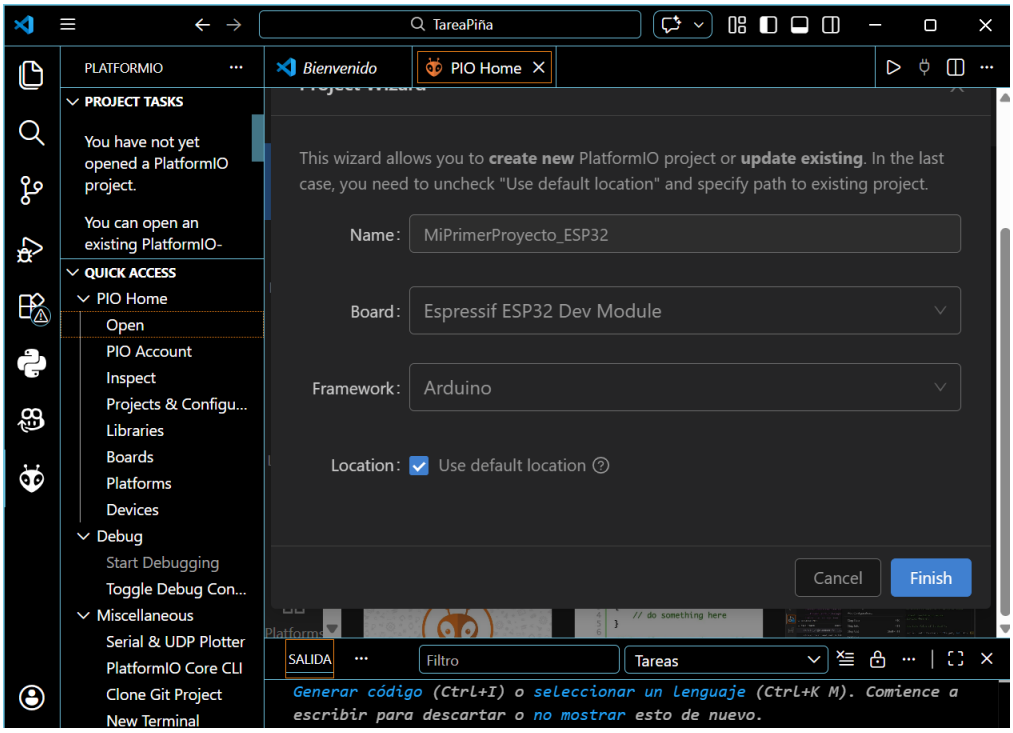


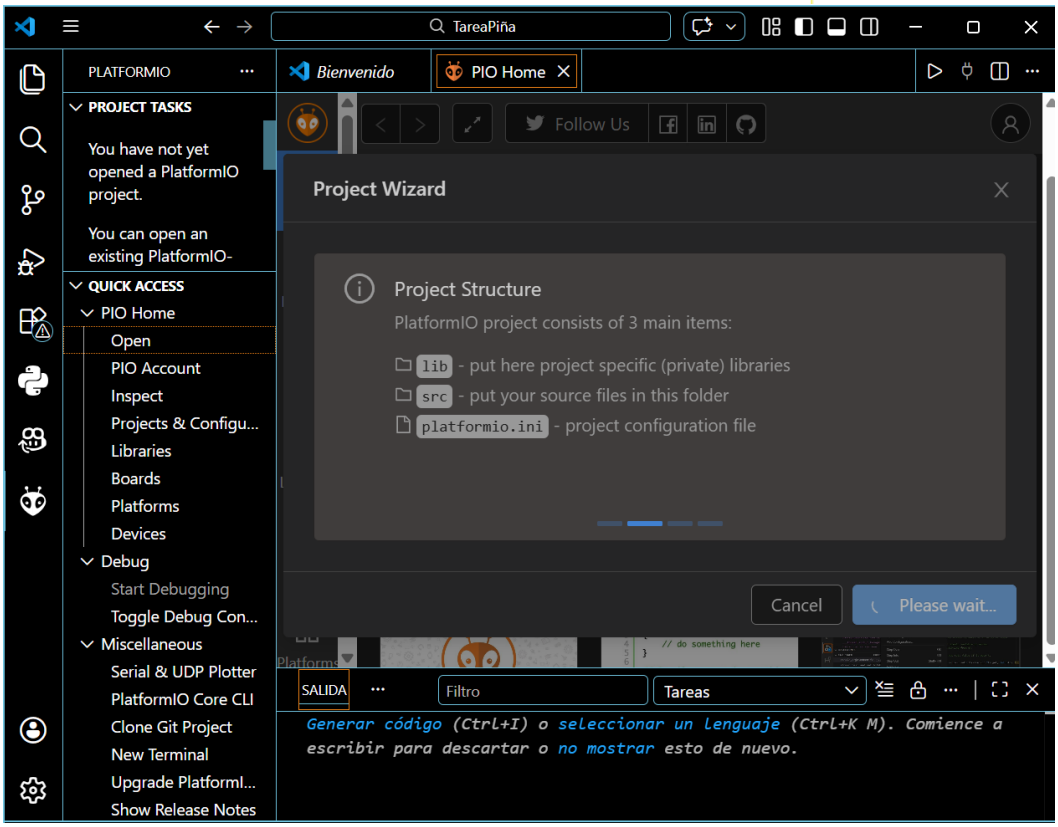
Descarga de su núcleo (core)

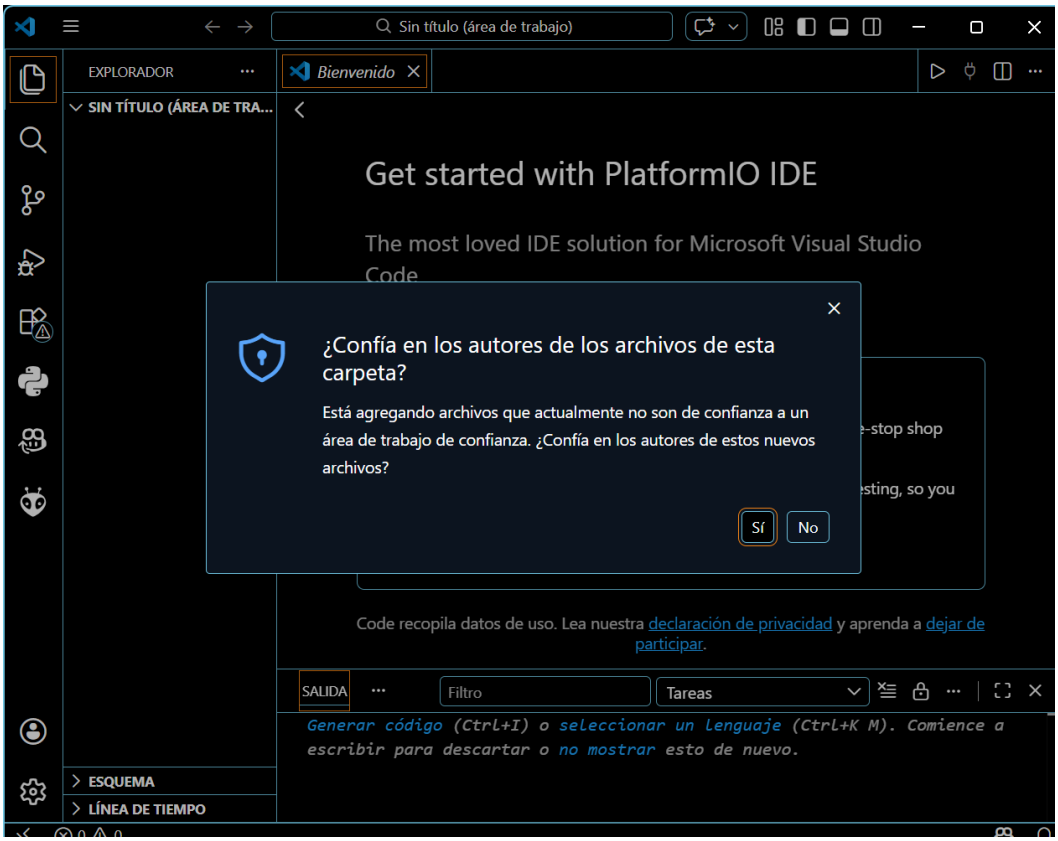


Crear el primer proyecto

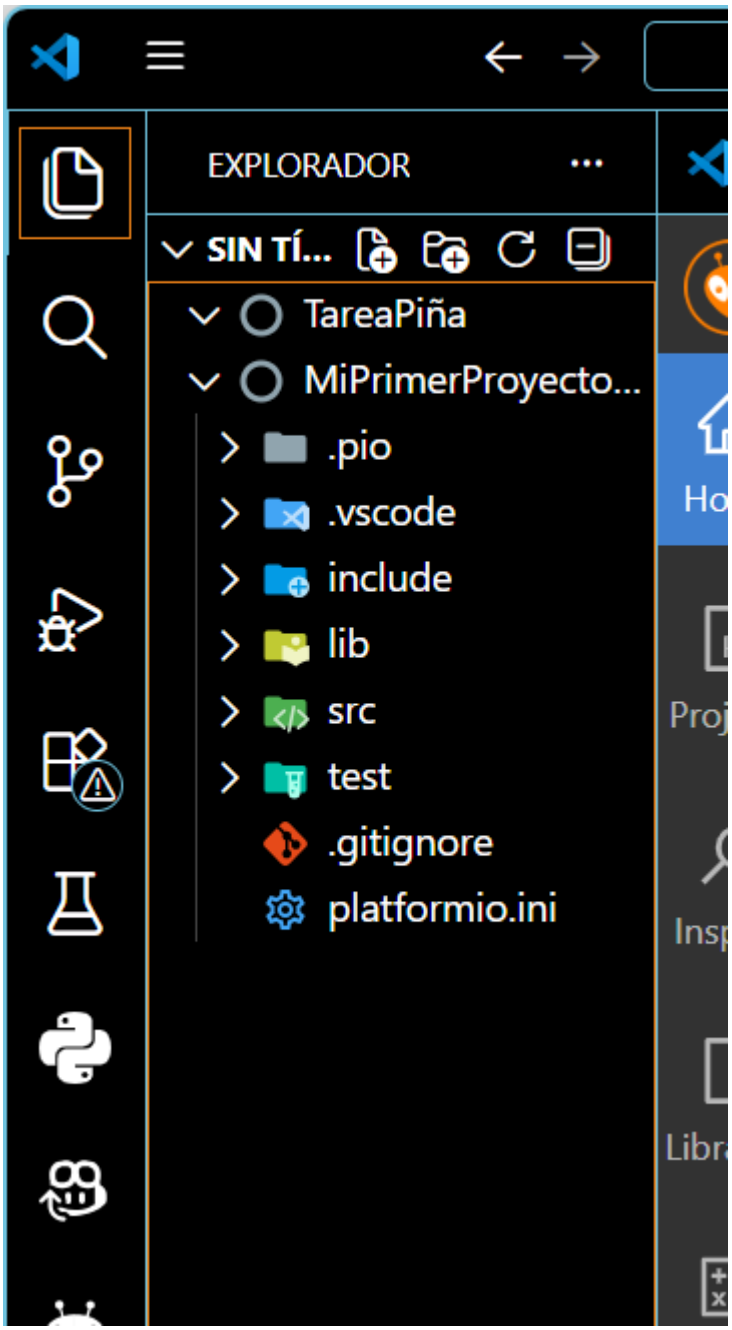




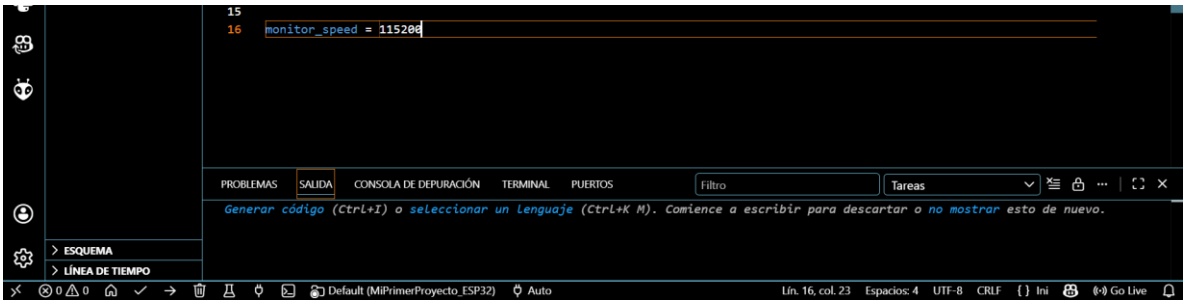

TareaPiña
PLATFORMIO
Bienvenido
PIO Home
PROJECT TASKS
You have not yet opened a PlatformIO project.
You can open an existing PlatformIO-
QUICK ACCESS
PIO Home
Open
PIO Account
Inspect
Projects & Configu...
Libraries
Boards
Platforms
Devices
Debug
Start Debugging
Toggle Debug Con...
Miscellaneous
Serial & UDP Plotter
PlatformIO Core CLI
Clone Git Project
New Terminal
Upgrade PlatformI...
Show Release Notes
Follow Us
Project Wizard
Project Structure
PlatformIO project consists of 3 main items:
lib - put here project specific (private) libraries
src - put your source files in this folder
platformio.ini - project configuration file
Cancel
Please wait...
SALIDA
Filtro
Tareas
Generar código (Ctrl+I) o seleccionar un lenguaje (Ctrl+K M). Comience a escribir para descartar o no mostrar esto de nuevo.


Sin título (área de trabajo)
EXPLORADOR
Bienvenido
SIN TÍTULO (ÁREA DE TRA...
Get started with PlatformIO IDE
The most loved IDE solution for Microsoft Visual Studio Code
¿Confía en los autores de los archivos de esta carpeta?
Está agregando archivos que actualmente no son de confianza a un área de trabajo de confianza. ¿Confía en los autores de estos nuevos archivos?
Sí
No
Code recopila datos de uso. Lea nuestra declaración de privacidad y aprenda a dejar de participar.
SALIDA
Filtro
Tareas
Generar código (Ctrl+I) o seleccionar un lenguaje (Ctrl+K M). Comience a escribir para descartar o no mostrar esto de nuevo.
ESQUEMA
LÍNEA DE TIEMPO

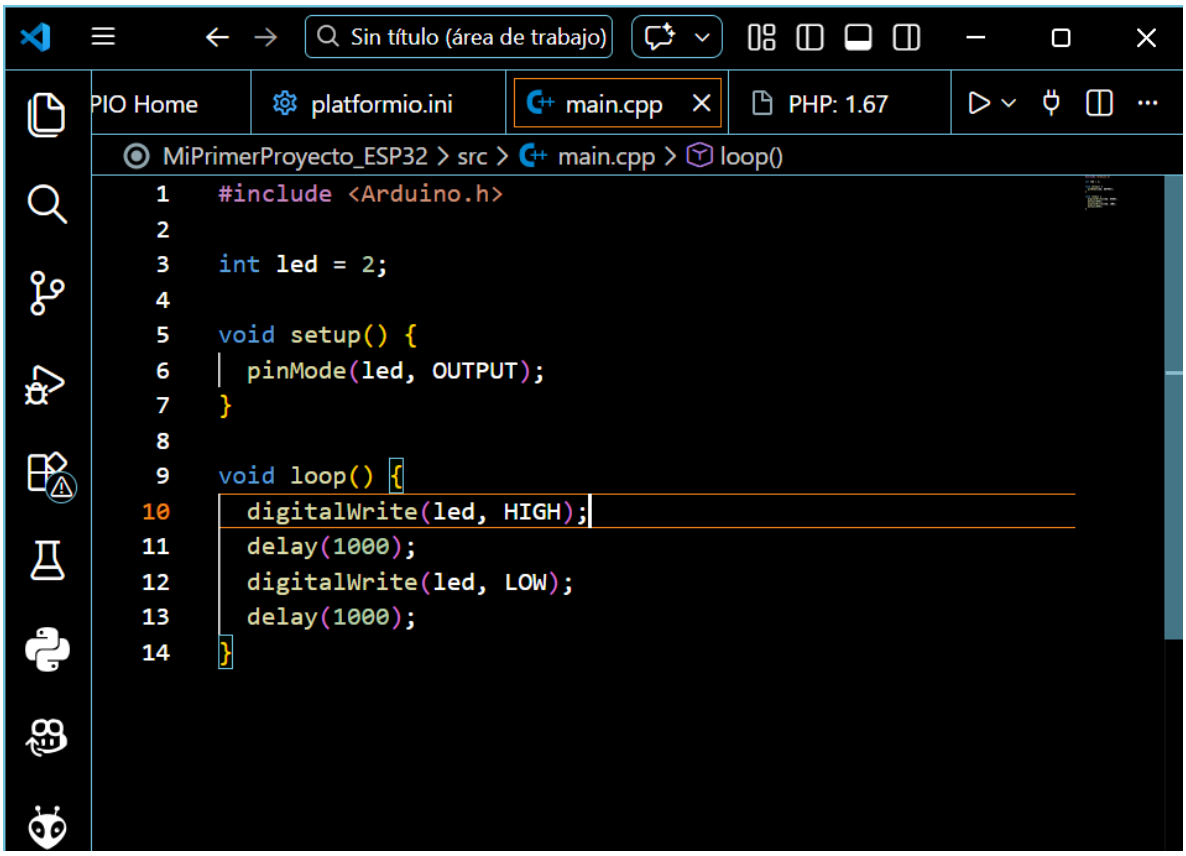
Anatomía de un proyecto profesional

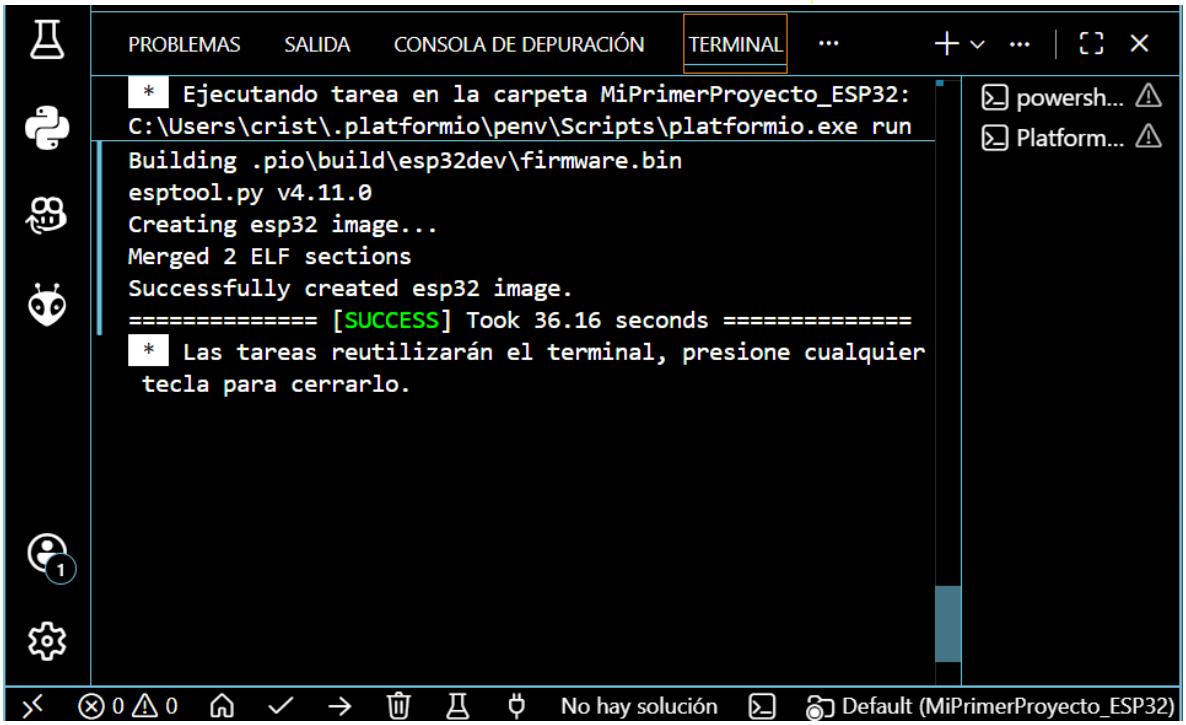


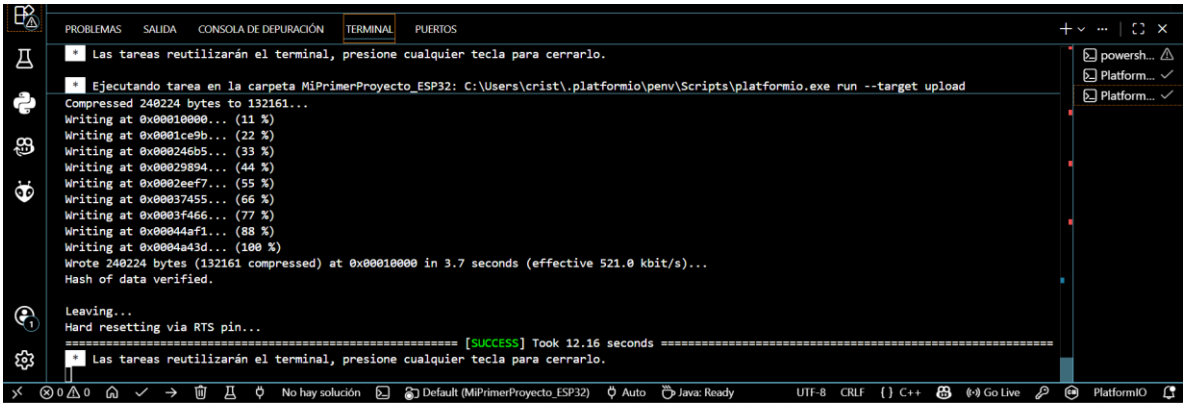
Compilar y subir el código a hardware



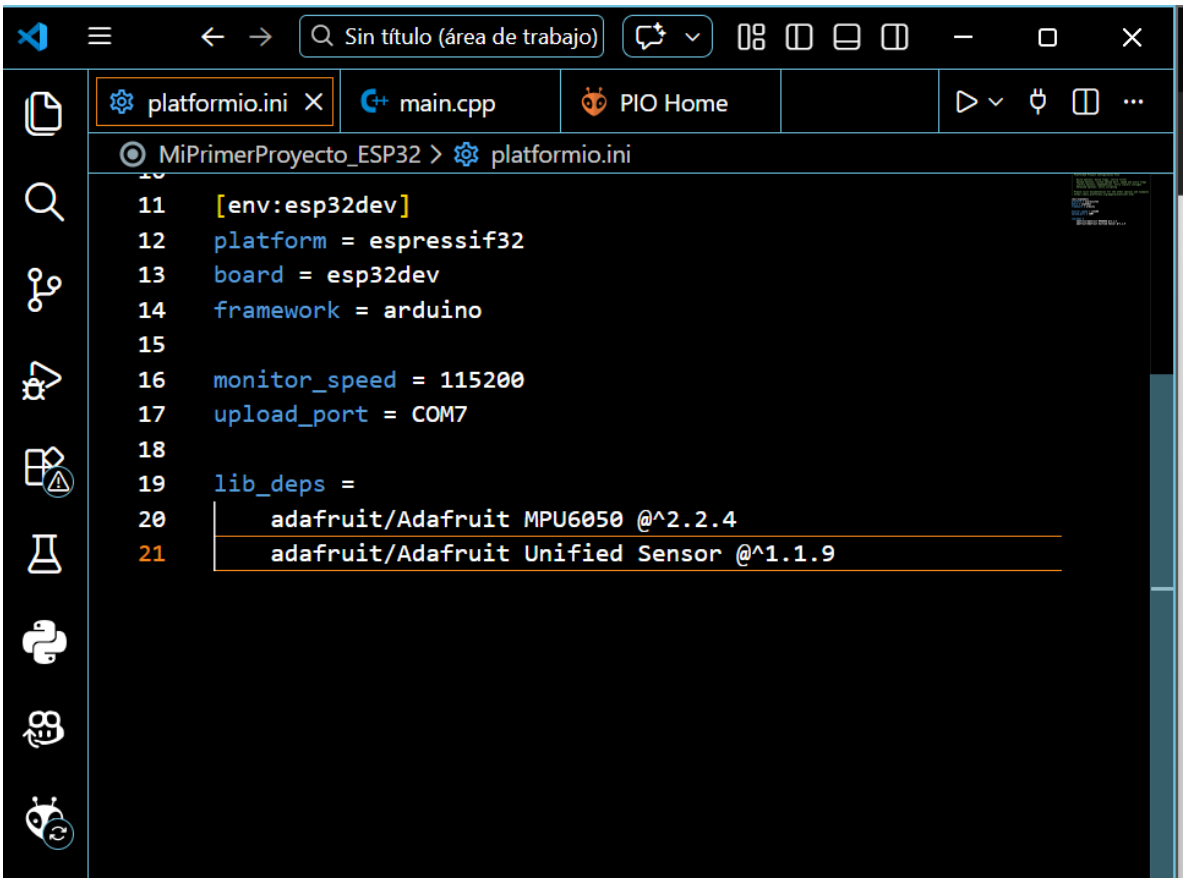
Código de prueba

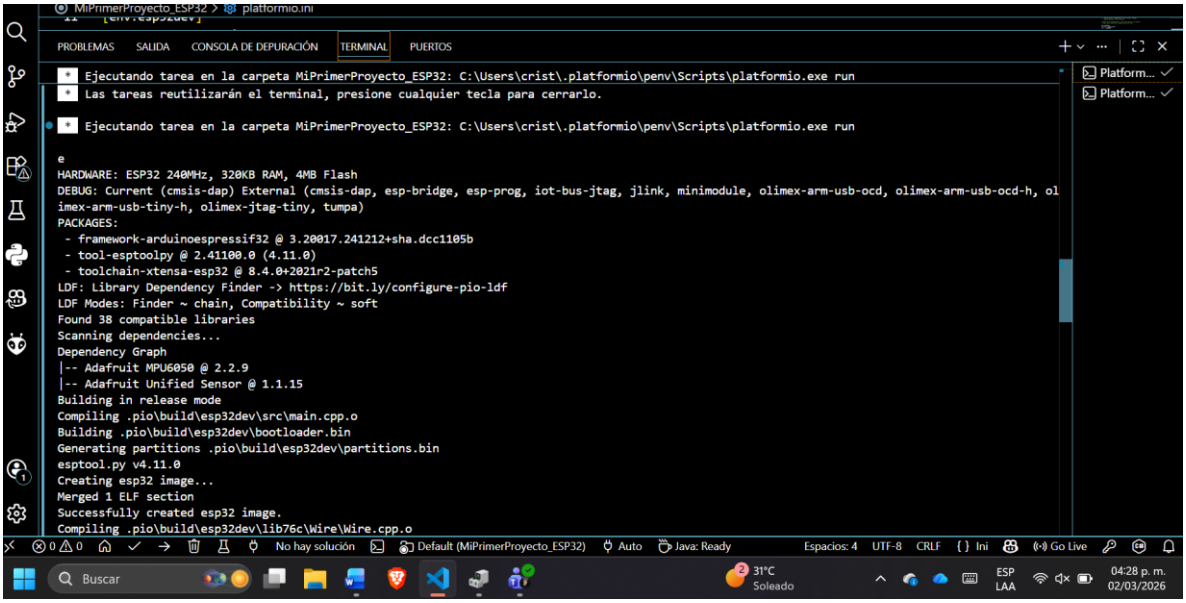



PROBLEMAS
SALIDA
CONSOLA DE DEPURACIÓN
TERMINAL
* Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32:
C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
Building .pio\build\esp32dev\firmware.bin
esptool.py v4.11.0
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
========== [SUCCESS] Took 36.16 seconds ==========
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
powersh...
Platform...
No hay solución
Default (MiPrimerProyecto_ESP32)


PROBLEMAS
SALIDA
CONSOLA DE DEPURACIÓN
TERMINAL
PUERTOS
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
* Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run --target upload
Compressed 240224 bytes to 132161...
Writing at 0x00010000... (11 %)
Writing at 0x0001ce9b... (22 %)
Writing at 0x000246b5... (33 %)
Writing at 0x00029894... (44 %)
Writing at 0x0002eef7... (55 %)
Writing at 0x00037455... (66 %)
Writing at 0x0003f466... (77 %)
Writing at 0x00044af1... (88 %)
Writing at 0x0004a43d... (100 %)
Wrote 240224 bytes (132161 compressed) at 0x00010000 in 3.7 seconds (effective 521.0 kbit/s)...
Hash of data verified.
Leaving...
Hard resetting via RTS pin...
========== [SUCCESS] Took 12.16 seconds ==========
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
powersh...
Platform...
Platform...
No hay solución
Default (MiPrimerProyecto_ESP32)
Auto
Java: Ready
UTF-8
CRLF
C++
Go Live
PlatformIO

Librerías

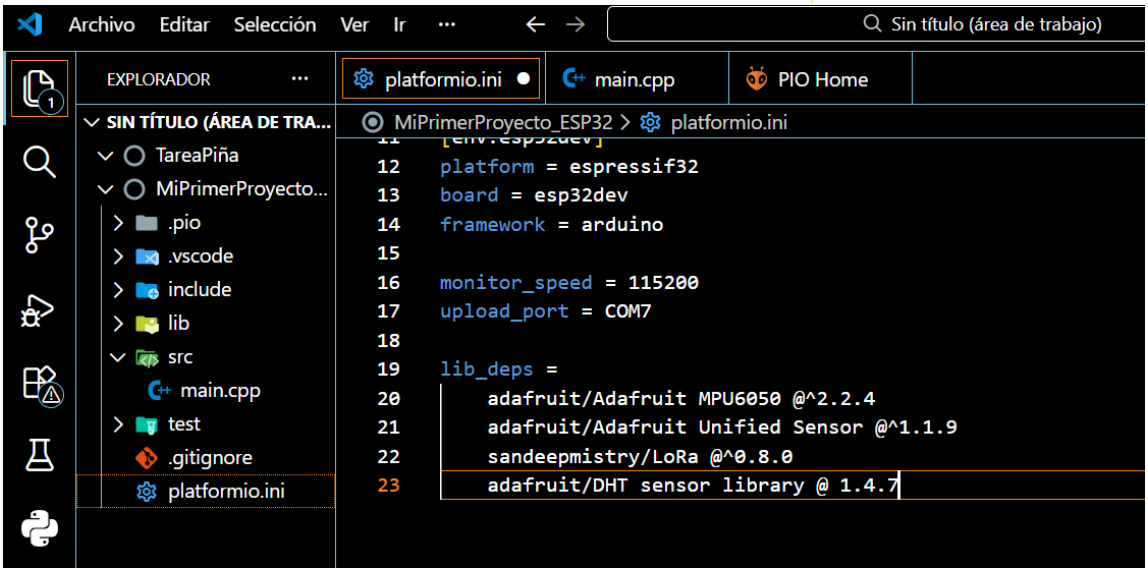

Sin título (área de trabajo)
platformio.ini
main.cpp
PIO Home
MiPrimerProyecto_ESP32 > platformio.ini
[env:esp32dev]
platform = espressif32
board = esp32dev
framework = arduino
monitor_speed = 115200
upload_port = COM7
lib_deps =
adafruit/Adafruit MPU6050 @^2.2.4
adafruit/Adafruit Unified Sensor @^1.1.9

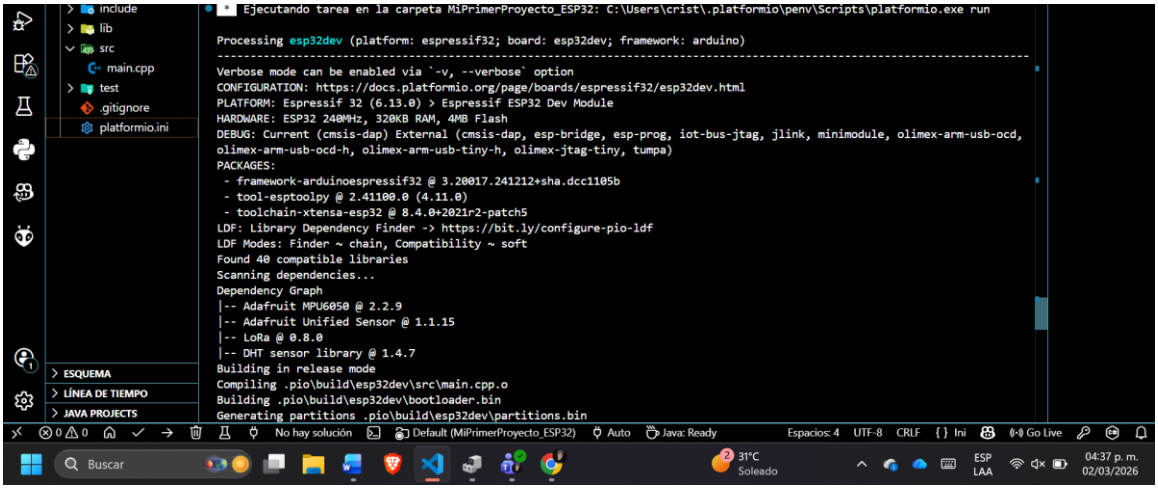

PROBLEMAS
SALIDA
CONSOLA DE DEPURACIÓN
TERMINAL
PUERTOS
Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
e
HARDWARE: ESP32 240MHz, 320KB RAM, 4MB Flash
DEBUG: Current (cmsis-dap) External (cmsis-dap, esp-bridge, esp-prog, iot-bus-jtag, jlink, minimodule, olimex-arm-usb-ocd, olimex-arm-usb-ocd-h, ol
imex-arm-usb-tiny-h, olimex-jtag-tiny, tumpa)
PACKAGES:
- framework-arduinoespressif32 @ 3.20017.241212+sha.dcc1105b
- tool-esptoolpy @ 2.41100.0 (4.11.0)
- toolchain-xtensa-esp32 @ 8.4.0+2021r2-patch5
LDF: Library Dependency Finder -> https://bit.ly/configure-pio-ldf
LDF Modes: Finder ~ chain, Compatibility ~ soft
Found 38 compatible libraries
Scanning dependencies...
Dependency Graph
|-- Adafruit MPU6050 @ 2.2.9
|-- Adafruit Unified Sensor @ 1.1.15
Building in release mode
Compiling .pio\build\esp32dev\src\main.cpp.o
Building .pio\build\esp32dev\bootloader.bin
Generating partitions .pio\build\esp32dev\partitions.bin
esptool.py v4.11.0
Creating esp32 image...
Merged 1 ELF section
Successfully created esp32 image.
Compiling .pio\build\esp32dev\lib76c\Wire\Wire.cpp.o

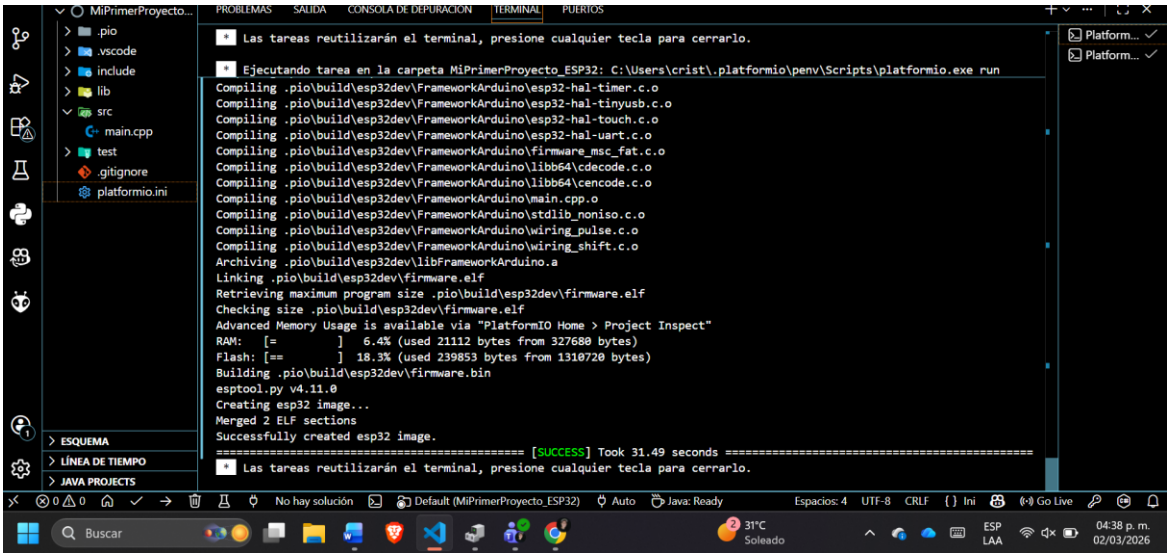
```
Archiving .pio\build\esp32dev\libFrameworkArduino.a
Linking .pio\build\esp32dev\firmware.elf
Retrieving maximum program size .pio\build\esp32dev\firmware.elf
Checking size .pio\build\esp32dev\firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM:   [=         ]   6.4% (used 21112 bytes from 327680 bytes)
Flash: [==        ]  18.3% (used 239853 bytes from 1310720 bytes)
Building .pio\build\esp32dev\firmware.bin
esptool.py v4.11.0
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
========================================================= [SUCCESS] Took 30.70 seconds =========================================================
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
```

```
PROBLEMAS   SALIDA   CONSOLA DE DEPURACIÓN   TERMINAL   PUERTOS
* Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.

* Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run

Processing esp32dev (platform: espressif32; board: esp32dev; framework: arduino)
-------------------------------------------------------------------------------
Verbose mode can be enabled via `-v, --verbose` option
CONFIGURATION: https://docs.platformio.org/page/boards/espressif32/esp32dev.html
PLATFORM: Espressif 32 (6.13.0) > Espressif ESP32 Dev Module
HARDWARE: ESP32 240MHz, 320KB RAM, 4MB Flash
DEBUG: Current (cmsis-dap) External (cmsis-dap, esp-bridge, esp-prog, iot-bus-jtag, jlink, minimodule, olimex-arm-usb-ocd, olimex-arm-usb-ocd-h, ol
imex-arm-usb-tiny-h, olimex-jtag-tiny, tumpa)
PACKAGES:
 - framework-arduinoespressif32 @ 3.20017.241212+sha.dcc1105b
 - tool-esptoolpy @ 2.41100.0 (4.11.0)
 - toolchain-xtensa-esp32 @ 8.4.0+2021r2-patch5
LDF: Library Dependency Finder -> https://bit.ly/configure-pio-ldf
LDF Modes: Finder ~ chain, Compatibility ~ soft
Found 39 compatible libraries
Scanning dependencies...
Dependency Graph
|-- Adafruit MPU6050 @ 2.2.9
|-- Adafruit Unified Sensor @ 1.1.15
|-- LoRa @ 0.8.0
Building in release mode
Compiling .pio\build\esp32dev\src\main.cpp.o
Building .pio\build\esp32dev\bootloader.bin
Generating partitions .pio\build\esp32dev\partitions.bin
```

```
PROBLEMAS   SALIDA   CONSOLA DE DEPURACIÓN   TERMINAL   PUERTOS
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.

* Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-timer.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-tinyusb.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-touch.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-uart.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\firmware_msc_fat.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\libb64\cdecode.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\libb64\cencode.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\main.cpp.o
Compiling .pio\build\esp32dev\FrameworkArduino\stdlib_noniso.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\wiring_pulse.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\wiring_shift.c.o
Archiving .pio\build\esp32dev\libFrameworkArduino.a
Linking .pio\build\esp32dev\firmware.elf
Retrieving maximum program size .pio\build\esp32dev\firmware.elf
Checking size .pio\build\esp32dev\firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM:   [=         ]   6.4% (used 21112 bytes from 327680 bytes)
Flash: [==        ]  18.3% (used 239853 bytes from 1310720 bytes)
Building .pio\build\esp32dev\firmware.bin
esptool.py v4.11.0
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
============== [SUCCESS] Took 30.81 seconds ==============
* Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
```


Archivo Editar Selección Ver Ir
Sin título (área de trabajo)
EXPLORADOR
platformio.ini
main.cpp
PIO Home
SIN TÍTULO (ÁREA DE TRA...
TareaPiña
MiPrimerProyecto...
.pio
.vscode
include
lib
src
main.cpp
test
.gitignore
platformio.ini
MiPrimerProyecto_ESP32 > platformio.ini
12 platform = espressif32
13 board = esp32dev
14 framework = arduino
16 monitor_speed = 115200
17 upload_port = COM7
19 lib_deps =
20 adafruit/Adafruit MPU6050 @^2.2.4
21 adafruit/Adafruit Unified Sensor @^1.1.9
22 sandeepmistry/LoRa @^0.8.0
23 adafruit/DHT sensor library @ 1.4.7


Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
Processing esp32dev (platform: espressif32; board: esp32dev; framework: arduino)
Verbose mode can be enabled via `-v, --verbose` option
CONFIGURATION: https://docs.platformio.org/page/boards/espressif32/esp32dev.html
PLATFORM: Espressif 32 (6.13.0) > Espressif ESP32 Dev Module
HARDWARE: ESP32 240MHz, 320KB RAM, 4MB Flash
DEBUG: Current (cmsis-dap) External (cmsis-dap, esp-bridge, esp-prog, iot-bus-jtag, jlink, minimodule, olimex-arm-usb-ocd, olimex-arm-usb-ocd-h, olimex-arm-usb-tiny-h, olimex-jtag-tiny, tumpa)
PACKAGES:
- framework-arduinoespressif32 @ 3.20017.241212+sha.dcc1105b
- tool-esptoolpy @ 2.41100.0 (4.11.0)
- toolchain-xtensa-esp32 @ 8.4.0+2021r2-patch5
LDF: Library Dependency Finder -> https://bit.ly/configure-pio-ldf
LDF Modes: Finder ~ chain, Compatibility ~ soft
Found 40 compatible libraries
Scanning dependencies...
Dependency Graph
|-- Adafruit MPU6050 @ 2.2.9
|-- Adafruit Unified Sensor @ 1.1.15
|-- LoRa @ 0.8.0
|-- DHT sensor library @ 1.4.7
Building in release mode
Compiling .pio\build\esp32dev\src\main.cpp.o
Building .pio\build\esp32dev\bootloader.bin
Generating partitions .pio\build\esp32dev\partitions.bin


PROBLEMAS SALIDA CONSOLA DE DEPURACIÓN TERMINAL PUERTOS
Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.
Ejecutando tarea en la carpeta MiPrimerProyecto_ESP32: C:\Users\crist\.platformio\penv\Scripts\platformio.exe run
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-timer.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-tinyusb.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-touch.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\esp32-hal-uart.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\firmware_msc_fat.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\libb64\cdecode.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\libb64\cencode.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\main.cpp.o
Compiling .pio\build\esp32dev\FrameworkArduino\stdlib_noniso.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\wiring_pulse.c.o
Compiling .pio\build\esp32dev\FrameworkArduino\wiring_shift.c.o
Archiving .pio\build\esp32dev\libFrameworkArduino.a
Linking .pio\build\esp32dev\firmware.elf
Retrieving maximum program size .pio\build\esp32dev\firmware.elf
Checking size .pio\build\esp32dev\firmware.elf
Advanced Memory Usage is available via "PlatformIO Home > Project Inspect"
RAM: [=] 6.4% (used 21112 bytes from 327680 bytes)
Flash: [==] 18.3% (used 239853 bytes from 1310720 bytes)
Building .pio\build\esp32dev\firmware.bin
esptool.py v4.11.0
Creating esp32 image...
Merged 2 ELF sections
Successfully created esp32 image.
========== [SUCCESS] Took 31.49 seconds ==========
Las tareas reutilizarán el terminal, presione cualquier tecla para cerrarlo.

Código de ejemplo en una ESP32